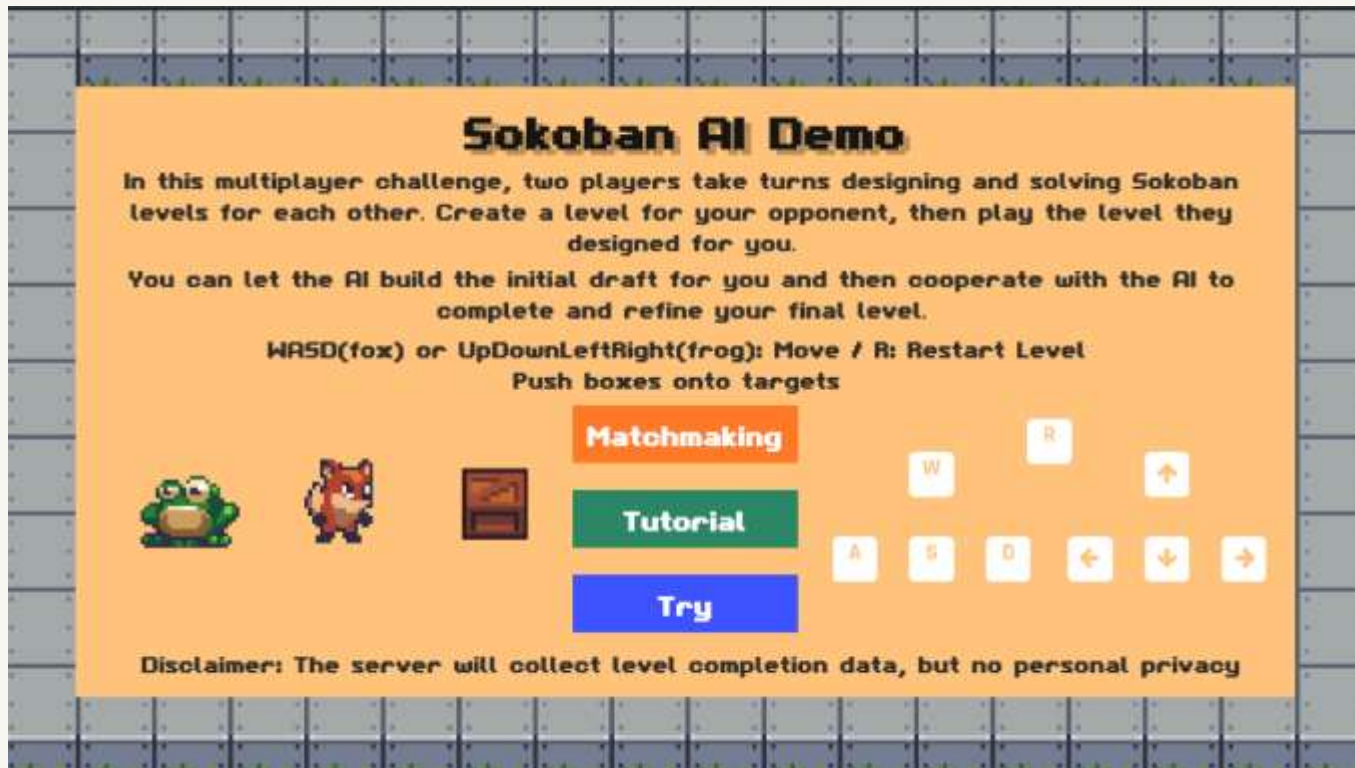


Sokoban AI Demo 教程 / Tutorial

多人共创推箱子挑战 | Bilingual Guide for Multiplayer Co-Creation



你将做什么？ / What will you do?

- 与另一位玩家互相设计并游玩 Sokoban 关卡。
/Design and play Sokoban levels with another player.
- 先创建或加入房间，再进入共创地图阶段。
/Create or join a room, then enter co-creation.
- 和 AI 一起修改、试玩、验证并确认最终关卡
/Revise, test, verify, and confirm the final level with AI.
- 最后游玩对手的关卡，并完成结果记录。
/Finally, play your opponent level and record results.



关键词 / Keywords:

狐狸 = 试玩自己的关卡 / Fox = test your own level

青蛙 = 游玩对手关卡 / Frog = opponent level

箱子 = 推到目标点 / Box = push to the target

1. 基础规则与操作 / Basic Rules & Controls

先认识角色、按键和时间限制 / Learn the characters, controls, and time limits first

角色与操作 / Roles & Controls

- 狐狸：试玩自己设计的关卡。
/ Fox: test your own level.
- 操作：W / A / S / D 移动，R 重新开始。
/ Move with W/A/S/D; press R to restart.
- 青蛙：游玩对手设计的关卡。
/ Frog: play your opponent level.
- 操作：方向键 ↑ ↓ ← → 移动，R 重新开始。
/ Move with arrow keys; press R to restart.
- 核心目标：把箱子推到目标点。
/ Goal: push boxes onto targets.



狐狸 / Fox
WASD



青蛙 / Frog
方向键 / Arrow keys



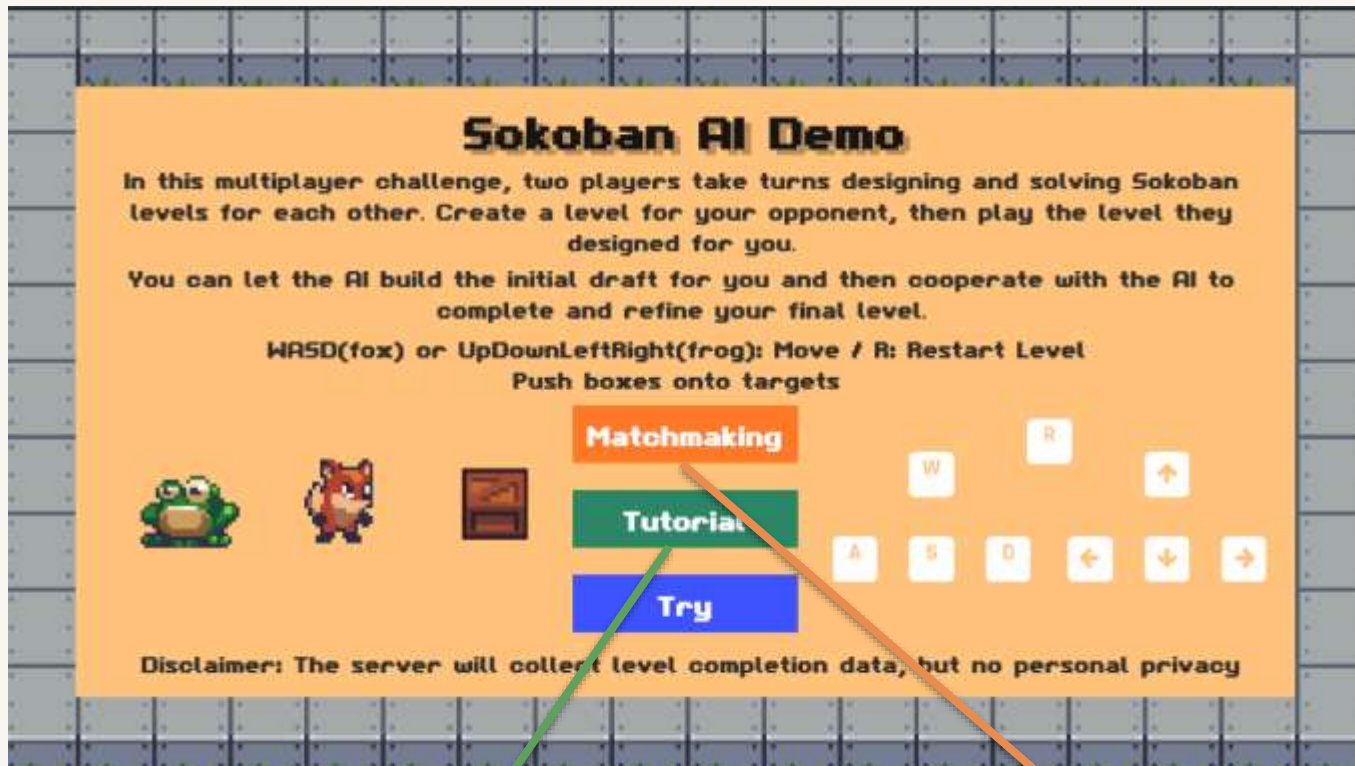
箱子 / Box
推到目标点 / Push to target

共创阶段限时 10 分钟
Co-creation limit: 10 min

游玩对手关卡限时 30 秒
Opponent level limit: 30 sec

2. 从主菜单开始 / Start from the Main Menu

阅读说明后，进入匹配流程 / After reading the instructions, enter matchmaking



教程按钮
Tutorial button

匹配入口
Matchmaking entry

主菜单说明 / Main Menu Guide

- Tutorial: 查看本教程。
/ View this tutorial.
- Server: 连接服务器。
/ Connect to the server.
- Matchmaking: 进入匹配大厅。
/ Enter the matching lobby.
- Try: 可进入本地试玩。
/ Try local practice.
- 底部说明: 系统会收集关卡完成数据, 但不采集个人隐私。
/ Footer note: completion data is collected, but personal privacy is not.

2A. 创建首版 Draft / Create the Starting Draft

回答四个中立问题，帮助 AI 理解你的初始设计方向 / Answer four neutral questions to help AI understand your initial design direction

Draft

Answer four quick questions. The AI will summarize your choices and give you a suggestion.
You can also adjust the parameter by yourself.

DESIGN BRIEF

How much inspection should the map require before the first move?

- Little inspection before acting
- Some inspection of boxes, goals, and passages
- Broader inspection and planning before acting
- No preference

AI reflection will appear here.

Layout < >

Difficulty < >

AI difficulty suggestion will appear here.

AI layout suggestion will appear here.

CONFIRM

调节参数
Adjust parameters

选择答案
Select answers

Draft 第一步 / Draft Step 1

- 阅读每个问题，选择最符合你想让对手获得的体验的选项。
/ Choose the option that best matches the experience you want your opponent to have.
- 这些回答只用于帮助 AI 生成初始建议，不是最终限制。
/ These answers guide the AI's first suggestion; they are not final limits.
- 确认前，你仍可以手动调整 Layout 与 Difficulty。
/ Before confirming, you can still adjust Layout and Difficulty manually.
- 没有明确想法时，可以选择 No preference。
/ If you have no clear preference, choose No preference.

2B. 创建首版 Draft / Create the Starting Draft

查看 AI 总结和参数建议，必要时手动调整 / Review the AI summary and parameter suggestions, then adjust if needed

Draft

Answer four quick questions. The AI will summarize your choices and give you a suggestion.
You can also adjust the parameter by yourself.

DESIGN BRIEF

AI is reviewing your initial-map preferences.

- Short routes connecting nearby decisions
- Mostly direct routes with some detours
- Longer routes with exploration or return paths
- No preference

AI reflection: Your choices suggest a quick opening move with pushes that depend on each other. It sounds like you want a few connected areas with longer routes between them.

Layout: Open

Difficulty: Medium

AI difficulty suggestion: Medium

AI layout suggestion: Open

CONFIRM

AI 总结与推荐
AI summary & suggestion

确认并继续
Confirm & continue

AI 推荐说明 / AI Recommendation Guide

- AI 会根据你的回答生成 reflection，总结它理解到的设计方向。
/ The AI generates a reflection summarizing its read of your design direction.
- 查看 AI 推荐的 Difficulty 和 Layout 是否符合你的想法。
/ Check whether the suggested Difficulty and Layout match your intention.
- 你可以接受建议，也可以手动调整参数。
/ You may accept the suggestion or adjust the parameters manually.
- 只有当首版方向合适时，再点击 CONFIRM。
/ Click CONFIRM only when the starting direction feels appropriate.
- 这只是首版草案，之后仍会进入 AI 共创修改阶段。
/ This is only the starting draft; you will still revise it with AI later.

3. 匹配大厅：先创建房间码，再加入房间 / Matchmaking: Create a Code, Then Join

一名玩家创建房间，另一名玩家使用房间码加入 / One player creates a room; the other joins with the code

01 CREATE

Create a room

Open a private match and share the generated code with your opponent.

ROOM CODE APPEARS HERE

CREATE ROOM

步骤 1：点击 CREATE ROOM
Step 1: Click CREATE ROOM

生成房间码并分享给对手
/ Share the room code with your opponent

02 JOIN

Join a room

Enter the six-digit room code shared by the room host.

JOIN ROOM

步骤 2：输入 6 位房间码
Step 2: Enter the 6-digit code

点击 JOIN ROOM 进入房间
/ Click JOIN ROOM to enter

注意：你们两位玩家需要进入同一个房间，随后才能开始后续共创与交换关卡。 / Note: both players must enter the same room before co-creation and level exchange can begin.

4. 共创阶段总览 / Co-Creation Stage Overview

与 AI 讨论并持续完善当前 Stage / Discuss with AI and refine the current Stage

5 Sokoban 共创实验室

可持续运营的关卡设计工作台

TIME LEFT 09:42

English

正在载入会话...

关卡历史

Stages

1

Stage 1

当前

初始版本

✓ 已验证可解

这个 Stage 暂时还没有可测试记录。

AI

共创对话

讨论并继续完善当前 Stage

描述生成

这个初始地图的整体骨架挺清楚的：左上角一个目标点，右下角一片水域把地图下半部分切成了两个区域，两个箱子都落在下半部分，玩家从底部出发。我比较在意的是 B2 和那片水的关系——B2 在 (6,5)，水域在右侧 (7-9, 8-9)，中间隔着几格地板，所以推 B2 往右时，路线会贴着水域边缘走，玩家需要留意别把箱子推进死角。另一个让我注意的是 T2 在 (6,11)，离水域很近，如果想把 B2 送到那里，最后一步的推法会很受限制，可能得从上方或左侧绕。整体上，这个布局看起来是让玩家先处理左边的箱子，再想办法把右边的箱子送过水域附近的窄道，我觉得这个方向挺有潜力的。

你可以先说说你的第一反应或试玩当前关卡，之后不满意的话，你可以指出来，我们一起探讨商量然后制定方案，或者你自己在右侧面板进行局部编辑；我只能协助进行小范围更改。提供可审查的改动内容以及帮助你梳理思路，较大的改动我建议由你亲来试一试。

STAGE 1

当前关卡

试玩这个 STAGE

地图

墙

水

玩家

箱子

目标

空白

正在编辑当前 Stage：只有明确保存后才会产生新版本。

✓ 已验证可解

15 步 · 16 次访问 · 9889 states

保存为新 STAGE

放弃草稿

试玩证据

这个 Stage 暂时还没有可测试记录。

确认

确认后请冻结编辑。

左侧：Stage 历史

Left: Stage History

中间：共创对话区

Middle: Co-Creation Chat

右侧：地图编辑与验证

Right: Map Editing & Verification

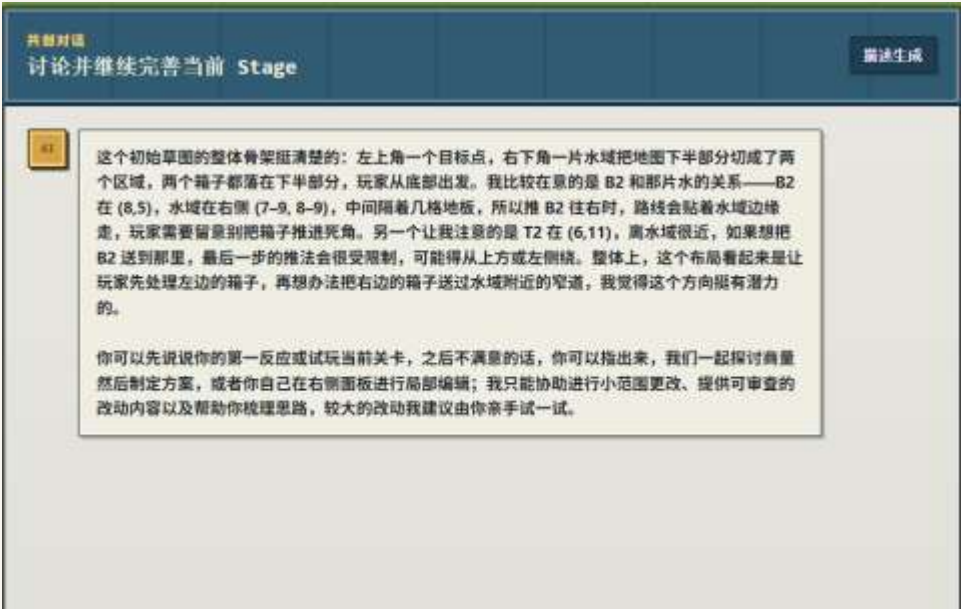
5. 共创界面怎么用 / How to Use the Co-Creation Interface

看懂三个核心区域 / Understand the three core areas



Stage 历史 / Stage History

- 查看版本。
/ View versions.
- 回看当前 Stage。
/ Review the current Stage.



共创对话区 / Co-Creation Chat

- 描述想法、表达不满意之处。
/ Describe ideas and concerns.
 - 让 AI 帮你分析、建议、继续修改。
/ Ask AI to analyze, suggest, and revise.
- AI 修改能力有限，建议手动修改。
/ AI edits are limited; manual edits are recommended.



地图编辑区 / Map Editing Area

- 查看并编辑当前关卡。
/ View and edit the current level.
- 放置地面、墙、水、玩家、箱子、目标等元素。
/ Place floor, wall, water, player, box, and target tiles.

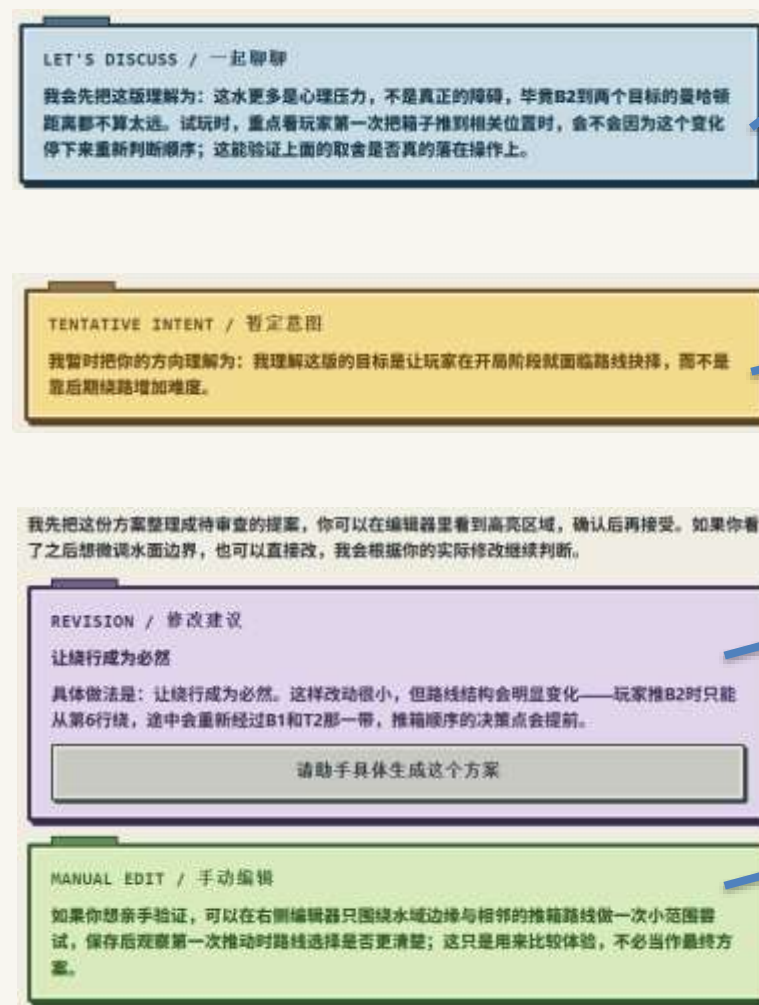
5A. 共创界面怎么用 / How to Use the Co-Creation Interface

看懂三个核心区域 / Understand the three core areas



共创对话区 / Co-Creation Chat

- 描述想法、表达不满意之处。
/ Describe ideas and concerns.
 - 让 AI 帮你分析、建议、继续修改。
/ Ask AI to analyze, suggest, and revise.
- AI 修改能力有限，建议手动修改。
/ Manual edits are recommended.



5B. 共创界面怎么用 / How to Use the Co-Creation Interface

看懂三个核心区域 / Understand the three core areas



地图编辑区 / Map Editing Area

- 查看并编辑当前关卡。
/ View and edit the current level.
- 放置地面、墙、水、玩家、箱子、目标等元素。
/ Place floor, wall, water, player, box, and target tiles.

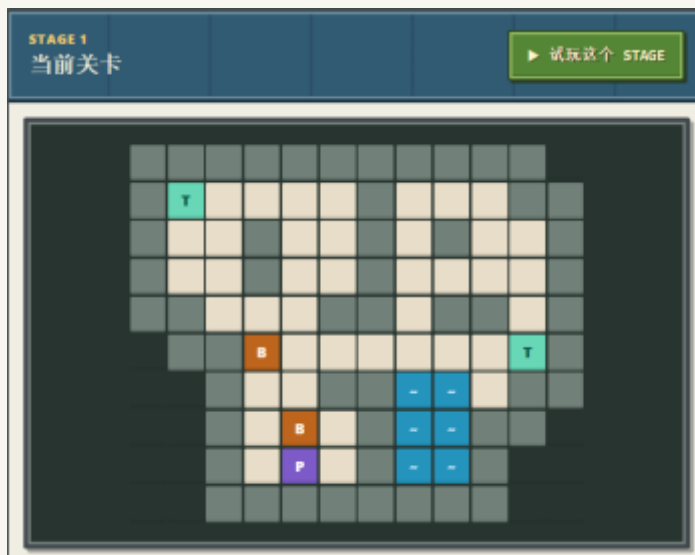
可选择的覆盖瓦片
Selectable replacement tiles

先选择要覆盖的瓦片作为画笔，再在图中点击替换瓦片
Choose a replacement tile as the brush, then click tiles on the map to replace them

修改后点击保存为新的 STAGE，再进入可解性检验
After editing, save as a new STAGE and run the solvability check

6. 试玩、验证与确认 / Playtest, Verify & Confirm

在确认前，请尽量自己体验和检查关卡 / Before confirming, try and check the level yourself



试玩当前 Stage / Playtest Current Stage

- 点击“试玩这个 STAGE”。
/ Click “Try this STAGE”.
- 试玩自己设计的关卡时，你操控狐狸。
/ When testing your own level, you control the fox.

验证与保存 / Verify & Save

- 系统会显示关卡是否可解。
/ The system shows whether the level is solvable.
- 可查看步数、推箱次数等信息。
/ Check steps and box-push counts.
- 满意后可保存为新 Stage。
/ Save as a new Stage when satisfied.

确认最终关卡 / Confirm Final Level

- 点击“确认最终 STAGE”。
/ Click “Confirm Final STAGE”.
- 确认后冻结编辑，并进入记录设计意图。
/ Editing freezes, then you record design intent.

7. 游玩对手的关卡 / Play Opponent Level

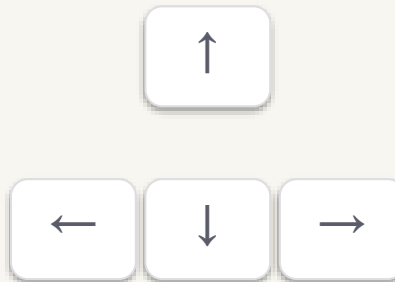
双方提交后，你将进入对手设计的关卡 / After both players submit, you enter your opponent level

挑战规则 / Challenge Rules

- 你将游玩另一位玩家设计的关卡。
/ You will play another player level.
- 此时可操控角色是青蛙。
/ You control the frog.
- 使用方向键 ↑ ↓ ← → 移动。
/ Move with the arrow keys.
- 本阶段限时 30 秒。
/ This stage is limited to 30 seconds.
- 目标仍然是把箱子推到目标点。
/ The goal is still to push boxes onto targets.



青蛙 = 对手关卡
Frog = opponent level



重要 / Important:
30 秒内尽力完成对手关卡
Try your best to finish within 30 seconds

8. 完整流程回顾 / Full Process Recap

开始前请再看一遍整体顺序 / Review the overall sequence before starting

1. 点击 Matchmaking并创建首版draft / Click Matchmaking and create draft

2. 创建房间码，另一位输入加入 / Create a room code; the other player joins

3. 与 AI 讨论并修改 Stage（10 分钟） / Discuss with AI and revise the Stage (10 min)

4. 试玩自己的关卡（狐狸 / WASD） / Test your own level (Fox / WASD)

5. 查看验证，保存新 Stage，并确认最终 STAGE / Verify, save, and confirm the final STAGE

6. 交换关卡并游玩对手关卡（青蛙 / 方向键 / 30 秒） / Swap and play opponent level (Frog / Arrow keys / 30 sec)

7. 完成记录与结果页面 / Complete the record and results page

祝你和队友顺利完成共创，并在挑战中获得有趣的体验！
Good luck co-creating with your teammate, and enjoy the challenge!